Huawei Access Node Hardware/Environmental Fault Resolution Procedure for Koramangala Datacenter

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8785

Description

The Huawei access node, designated as KOR-ACC-83, situated within the Koramangala Datacenter, has encountered a critical hardware/environmental fault, precipitating an ALARM_8785 warning. This failure domain, characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss, can be attributed to the intricate interplay between the node's hardware components and the ambient environmental conditions. The access node's role in facilitating communication between the datacenter and external networks underscores the importance of prompt resolution. The Huawei equipment, with its proprietary CLI dialect, necessitates a nuanced understanding of the underlying protocol and hardware mechanics. Specifically, the ASIC memory parity errors may arise from corrupted data or firmware glitches, while line card crashes could be indicative of overheating or power supply issues. The loss of power supply redundancy, in turn, compromises the node's ability to maintain uninterrupted service, thereby jeopardizing the overall network stability. A comprehensive resolution procedure, incorporating meticulous diagnostic and troubleshooting measures, is essential to mitigate these failures and ensure seamless network operation.

Pre-requisite Checks

- Dispatch remote hands to Koramangala Datacenter to perform physical checks
- Verify redundant link status to ensure failover mechanisms are operational
- Confirm environmental conditions (temperature, humidity) are within acceptable ranges

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to firmware glitches	High
Line card crashes resulting from overheating or power supply issues	Medium
Power supply redundancy loss due to hardware failure or configuration errors	Low

Diagnostic Commands & Expected Outputs

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Command: display device, Expected Output: Device information, including model, serial number, and firmware version, Description: This command provides an overview of the device's configuration, facilitating identification of potential hardware or firmware-related issues. By examining the device information, technicians can determine whether the fault lies in the hardware or software realm.

Command: display diagnostic-information, Expected Output: Detailed diagnostic information, including error logs and system statistics,
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Description: This command offers in-depth insights into the device's diagnostic information, enabling technicians to pinpoint specific error patterns or system anomalies that may be contributing to the failure domain. By analyzing the diagnostic information, technicians can develop a targeted approach to resolving the issue.

Command: display environment, Expected Output: Environmental parameters, including temperature, humidity, and power supply status, Description: This command allows technicians to assess the environmental conditions surrounding the device, which is crucial in identifying potential causes of hardware failures. By examining the environmental parameters, technicians can determine whether the fault is related to external factors, such as overheating or humidity.
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Resolution Steps

Step 1: Step: Update firmware to latest version, Command: startup system-software package://firmware.tar, Description: Updating the firmware to the latest version ensures that any known bugs or glitches are addressed, potentially resolving the ASIC memory parity errors and line card crashes. This step is crucial in maintaining the device's stability and performance.

Step 2: Step: Replace faulty line cards, Command: replace board 0/1, Description: Replacing faulty line cards is essential to prevent further crashes and ensure uninterrupted service. This step involves physically removing the faulty line card and installing a new one, which requires careful handling to avoid damaging surrounding components.

Step 3: Step: Restore power supply redundancy, Command: power-mode redundant, Description: Restoring power supply redundancy is vital to maintain the device's ability to provide uninterrupted service. This step involves configuring the power supply settings to ensure that the device can automatically switch to a redundant power source in case of a failure.

Post-Check Verification

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Command: display device, Expected Output: Updated device information, including new firmware version, Description: This command allows technicians to verify that the firmware update was successful and that the device is now running with the latest version. By examining the device information, technicians can confirm that the update has resolved any firmware-related issues.

Command: display diagnostic-information, Expected Output: Updated diagnostic information, indicating resolution of error logs and system statistics, Description: This command enables technicians to verify that the diagnostic information has been updated and that any error logs or system anomalies have been resolved. By analyzing the diagnostic information, technicians can confirm that the issue has been fully addressed.

Command: display environment, Expected Output: Updated environmental parameters, indicating stable temperature, humidity, and power supply status, Description: This command allows technicians to verify that the environmental conditions surrounding the device are stable and within acceptable ranges. By examining the environmental parameters, technicians can confirm that the device is operating in a suitable environment.
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Roll-back Procedures

- Step: Revert to previous firmware version, Command: startup system-software package://previous_firmware.tar, Description: If the updated firmware version introduces new issues or fails to resolve the existing problems, reverting to the previous firmware version may be necessary. This step involves downgrading the firmware to a previous version, which requires careful handling to avoid causing further instability.

- Step: Reinstall original line cards, Command: replace board 0/1 with original line card, Description: If the replacement line cards fail to resolve the issue or introduce new problems, reinstalling the original line cards may be necessary. This step involves physically removing the replacement line cards and reinstalling the original ones, which requires careful handling to avoid damaging surrounding components.

- Step: Restore original power supply configuration, Command: power-mode single, Description: If the restored power supply redundancy fails to provide uninterrupted service or introduces new issues, restoring the original power supply configuration may be necessary. This step involves configuring the power supply settings to their original state, which requires careful handling to avoid causing further instability.